Data Analytics using Python

Week 4 Question Bank Answer

1 What is Hypothesis Testing? Define Null hypothesis and Alternate hypothesis.

Ans.1) Hypothesis Testing is a statistical method used to determine whether a statement about the value of a population parameter should or should not be rejected based on sample data. It involves formulating two competing statements, the null hypothesis (H0) and the alternative hypothesis (Ha) and using data from a sample to test these hypotheses.

The null hypothesis, denoted by H0, is a tentative assumption about a population parameter. It is typically a statement of no effect or no difference and is often the hypothesis that a researcher aims to disprove.

The alternative hypothesis, denoted by Ha, is the opposite of what is stated in the null hypothesis. It is the statement that a researcher hopes to support with evidence from the sample data.

In some cases, it is easier to identify the alternative hypothesis first, while in other cases the null hypothesis is easier. The context of the situation is very important in determining how the hypotheses should be stated.

For example, in a study to test whether a new manufacturing method is better than the current method, the alternative hypothesis might be "The new manufacturing method is better," while the null hypothesis would be "The new method is no better than the old method."

In general, a hypothesis test about the value of a population mean μ must take one of the following three forms:

One-tailed (lower-tail): H0: $\mu \geq \mu_0$, Ha: $\mu < \mu_0$

One-tailed (upper-tail): H0: $\mu \leq \mu_0$, Ha: $\mu > \mu_0$

Two-tailed: H0: $\mu = \mu_0$, Ha: $\mu \neq \mu_0$

where μ_0 is the hypothesized value of the population mean.

Hypothesis testing is used to determine whether the sample data provide sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis. This is done by calculating a test statistic and comparing it to a critical value or a p-value. If the test statistic falls in the rejection region, the null hypothesis is rejected in favor of the alternative hypothesis. If the test statistic does not fall in the rejection region, the null hypothesis is not rejected.

It is important to note that hypothesis testing does not prove the alternative hypothesis to be true, but rather provides evidence against the null hypothesis.

2 Differentiate between type I and type II error.

Ans.2) Type I and Type II errors are two possible errors that can occur in hypothesis testing.

Type I error is rejecting the null hypothesis when it is actually true. This is also known as a false positive and is denoted by α. The probability of making a Type I error is called the level of significance of the test.

Type II error is accepting the null hypothesis when it is actually false. This is also known as a false negative and is denoted by β. The probability of making a Type II error is called the beta error.

The following table summarizes the differences between Type I and Type II errors:

	Null Hypothesis (H0) is true	Null Hypothesis (H0) is false
Correct Decision	Accept H0	Reject H0
Type I Error	Reject H0	Accept H0
Type II Error	Accept H0	Reject H0

	Type I Error (α)	Type II Error (β)
Definition	Rejecting H0 when it is true	Accepting H0 when it is false
Probability	Level of significance (α)	Beta error (β)
Consequence	False positive	False negative
Example	Convicting an innocent person	Failing to detect a defective product

In summary, Type I error is rejecting a true null hypothesis, while Type II error is accepting a false null hypothesis. The level of significance (α) is the probability of making a Type I error, while the beta error (β) is the probability of making a Type II error. The consequences of these errors can be significant, and it is important to carefully consider the potential risks and costs associated with each type of error.

3 Give the three Approaches for Hypothesis Testing.

Ans.3) The three approaches for hypothesis testing are:

1. **P-Value Approach**:
   - The p-value is the probability, computed using the test statistic, that measures the support or lack of support provided by the sample for the null hypothesis.
   - If the p-value is less than or equal to the level of significance α, the null hypothesis is rejected.
   - The rejection rule is to reject H0 if the p-value is less than α.

2. **Critical Value Approach**:
   - The critical value approach involves using critical values from a standard normal distribution or t-distribution to determine the rejection region.
   - The test statistic is compared to the critical value to decide whether to reject the null hypothesis.
   - The rejection rule is based on comparing the test statistic to the critical value.

3. **Confidence Interval Approach**:
   - In this approach, a confidence interval is constructed for the population parameter of interest.
   - If the hypothesized value falls within the confidence interval, the null hypothesis is not rejected.
   - If the hypothesized value is outside the confidence interval, the null hypothesis is rejected.

These approaches provide different ways to assess the evidence from sample data and make decisions regarding the null hypothesis in hypothesis testing.

4 What are the steps of Hypothesis Testing using P value approach?

Ans.4) The steps of hypothesis testing using the p-value approach are as follows:

1. Develop the null and alternative hypotheses: The null hypothesis (H0) is a tentative assumption about a population parameter, while the alternative hypothesis (Ha) is the opposite of what is stated in the null hypothesis.

2. Specify the level of significance (α): The level of significance is the probability of rejecting the null hypothesis when it is true.

3. Collect the sample data and compute the test statistic: The test statistic is a measure of how far the sample data is from the hypothesized value of the population parameter.

4. Compute the p-value: The p-value is the probability of observing a test statistic as extreme or more extreme than the one calculated, given that the null hypothesis is true.

5. Compare the p-value to the level of significance (α): If the p-value is less than the level of significance (α), reject the null hypothesis in favor of the alternative hypothesis.

6. Interpret the results: Based on the decision made in step 5, interpret the results in the context of the problem.

It is important to note that the p-value is a measure of evidence against the null hypothesis, and a small p-value indicates strong evidence against the null hypothesis. A p-value greater than the level of significance (α) does not prove the null hypothesis to be true, but rather indicates a lack of evidence to reject it.

5 What are the steps of Hypothesis Testing using critical value approach?

Ans.5) The steps of hypothesis testing using the critical value approach are as follows:

1. **Step 1: Develop the null and alternative hypotheses**: The null hypothesis (H0) is a tentative assumption about a population parameter, while the alternative hypothesis (Ha) is the opposite of what is stated in the null hypothesis.

2. **Step 2: Specify the level of significance (α)**: The level of significance is the probability of rejecting the null hypothesis when it is true.

3. **Step 3: Collect the sample data and compute the test statistic**: The test statistic is a measure of how far the sample data is from the hypothesized value of the population parameter.

4. **Step 4: Determine the critical value and rejection rule**: The critical value is the value of the test statistic that establishes the boundary of the rejection region. The rejection rule is based on the critical value and the direction of the alternative hypothesis.

5. **Step 5: Use the value of the test statistic and the rejection rule to determine whether to reject H0**: If the test statistic falls in the rejection region, reject the null hypothesis in favour of the alternative hypothesis.

6. **Step 6: Interpret the results**: Based on the decision made in step 5, interpret the results in the context of the problem.

In the critical value approach, the critical value is used to determine the rejection region, and the decision to reject or not reject the null hypothesis is based on whether the test statistic falls in the rejection region. This approach is often used when the p-value is not readily available or when the researcher wants to control the probability of making a Type I error.

6 The quality-control manager at a Li-BATTERY factory needs to determine whether the mean life of a large shipment of Li-Battery is equal to the specified value of 375 hours. The process standard deviation is known to be 100 hours. A random sample of 64 batteries indicates a sample mean life of 350 hours. State the null hypotheses.

Ans.6) The null hypothesis for the quality-control manager at the Li-BATTERY factory, who needs to determine whether the mean life of a large shipment of Li-Battery is equal to the specified value of 375 hours, would be:

Null Hypothesis (H0): The mean life of the Li-Battery shipment is equal to 375 hours.

In this scenario, the null hypothesis typically assumes no difference or no effect, stating that the mean life of the batteries is as specified. The quality-control manager will test this null hypothesis against an alternative hypothesis to determine if there is sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis.

7 In question 6, At the alpha = 0.05 level of significance, is there any evidence that the mean life differs from 375 hours.

Ans.7) To answer the question of whether there is evidence that the mean life of the Li-Battery differs from 375 hours at the alpha = 0.05 level of significance, we need to perform a hypothesis test using the critical value approach.

The null hypothesis (H0) is that the mean life of the Li-Battery is equal to 375 hours, and the alternative hypothesis (Ha) is that the mean life of the Li-Battery is not equal to 375 hours.

The test statistic for this hypothesis test is:

t = (sample mean - hypothesized mean) / (standard deviation / sqrt(sample size))

t = (350 - 375) / (100 / sqrt(64))

t = -4.9

The critical value for a two-tailed test with alpha = 0.05 and degrees of freedom

(df) = 64 - 1 = 63 is ±2.00.

Since the calculated test statistic (-4.9) is outside the critical value range (-2.00, 2.00), we reject the null hypothesis and conclude that there is evidence that the mean life of the Li-Battery differs from 375 hours at the alpha = 0.05 level of significance.

8 In question 6, compute the p-value ?

Ans.8) To compute the p-value for the hypothesis test in question 6, where the null hypothesis is that the mean life of the Li-Battery is equal to 375 hours (H0: $\mu = 375$), and the alternative hypothesis is that the mean life of the Li-Battery is different from 375 hours (Ha: $\mu \neq 375$), we can use the two-tailed p-value approach.

The test statistic for this hypothesis test is:

t = (sample mean - hypothesized mean) / (standard deviation / sqrt(sample size))

t = (350 - 375) / (100 / sqrt(64))

t = -4.9

The p-value is the probability of observing a test statistic as extreme or more extreme than the one calculated, given that the null hypothesis is true.

To find the p-value, we need to find the probability of observing a t-value as extreme or more extreme than -4.9, in both the lower and upper tails of the t-distribution with 63 degrees of freedom (df = 64 - 1).

Using a t-distribution table or a statistical software, we can find the two-tailed p-value for t = -4.9 and df = 63 to be approximately 0.00001 (or 0.001%).

Therefore, the p-value is 0.00001 (or 0.001%). This is a very small p-value, indicating that there is strong evidence against the null hypothesis, and we can reject the null hypothesis at any reasonable level of significance.

9 In question 6, at a 95% confidence interval, calculate the estimate of the population mean life of the battery.

Ans.9) To calculate the 95% confidence interval for the population mean life of the battery, we need to first find the margin of error (ME).

The formula for the margin of error for a population mean with a known standard deviation is:

ME = z*(σ/√n)

where

z is the z-score corresponding to the desired confidence level,

σ is the population standard deviation, and

n is the sample size.

For a 95% confidence interval, the z-score is 1.96.

Substituting the given values, we get:

ME = 1.96*(100/√64) = 24.5

The 95% confidence interval for the population mean life of the battery is:

(350 - 24.5, 350 + 24.5) = (325.5, 374.5)

Therefore, at a 95% confidence interval, the estimate of the population mean life of the battery is between 325.5 and 374.5 hours.

10 The mean cost of a hotel room in a city is \$168 per night. A random sample of 25 hotels resulted in X-bar = \$172.50 and sample standard deviation s = 15.40.

Calculate the t statistic.

Ans.10) To calculate the t statistic for the given situation, we will use the formula:

$t = (\bar{X} - \mu) / (s / \text{sqrt}(n))$

where

$\bar{X}$ is the sample mean,

μ is the population mean,

s is the sample standard deviation, and

n is the sample size.

Substituting the given values, we get:

$t = (172.50 - 168) / (15.40 / \text{sqrt}(25))$

$t = 1.25 / (15.40 / 5)$

$t = 1.25 / 3.08$

$t \approx 0.405$

Therefore, the t statistic for the given situation is approximately 0.405.

11 At the $\alpha = 0.05$ level. In question 10, what is the decision on the null hypothesis ?

Ans.11) To test the hypothesis that the mean cost of a hotel room in the city is $168 per night, we can use a one-sample t-test with a significance level of $\alpha = 0.05$.

Null hypothesis (H0): The mean cost of a hotel room in the city is $168 per night, or $\mu = 168$.

Alternative hypothesis (Ha): The mean cost of a hotel room in the city is different from $168 per night, or $\mu \neq 168$.

Using the given data, we can calculate the sample mean and sample standard deviation as:

Sample mean ($\bar{X}$): $172.50

Sample standard deviation (s): 15.40

Sample size (n): 25

We can then calculate the t-statistic as:

$t = (\bar{X} - \mu) / (s / \mathrm{sqrt}(n))$

$t = (172.50 - 168) / (15.40 / \mathrm{sqrt}(25))$

$t \approx 2.07$

Using a two-tailed test with $\alpha = 0.05$ and degrees of freedom (df) = n - 1 = 24, the critical t-values are ±2.064.

Since the calculated t-statistic (2.07) is greater than the critical t-value (2.064), we reject the null hypothesis and conclude that there is evidence that the mean cost of a hotel room in the city is different from $168 per night at the $\alpha = 0.05$ level of significance.

Therefore, the decision on the null hypothesis is to reject it in favor of the alternative hypothesis, indicating that there is a statistically significant difference between the sample mean cost of a hotel room and the hypothesized population mean cost of $168 per night.